

Dr. Meena Rao

Ph.D., M.Tech., B.E. | Associate Professor

Department of Electronics & Communication Engineering
Maharaja Surajmal Institute of Technology (MSIT), New Delhi
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PROFESSIONAL SUMMARY

- Extensively involved in university-level teaching and advanced academic research for over 21 years, featuring a strong publication track record in highly reputed indexed journals.
- Holds an AI Certification (awarded with a Certificate of Merit) from IIT Kharagpur and TCS iON, along with specialized internship credentials in Machine Learning and multiple SWAYAM certifications.
- Serves actively as Co-convener and earlier Convener (Sep 2017 – June 2026), spearheading campus recruitment drives for B.Tech students and maintaining long-term corporate linkages with industry HR leaders.

AREAS OF EXPERTISE

Machine Learning • Artificial Intelligence • Internet of Things (IoT) • Mobile Ad-Hoc Networks (MANETs)

WORK EXPERIENCE (21+ YEARS)

Associate Professor | Department of Electronics & Communication Engineering

Maharaja Surajmal Institute of Technology, New Delhi | June 6, 2017 – Present (9 years, 3 months)

- Serving as Associate Professor in the ECE Department, delivering core and advanced engineering courses while driving departmental academic strategy and research initiatives.
- Leading student career outcomes as Placement Committee Convener by fostering key corporate partnerships, coordinating recruitment drives, and connecting with HR leaders across top firms.

Assistant Professor | Department of Electronics & Communication Engineering

Maharaja Surajmal Institute of Technology (Affiliated to GGSIPU), New Delhi | January 14, 2005 – June 5, 2017 (12 years, 5 months)

- Conducted undergraduate lectures, laboratory sessions, and curriculum development across key Electronics & Communication engineering disciplines for over a decade.

EDUCATIONAL QUALIFICATIONS

- Ph.D. in Electronics & Communication Engineering (2016)
Gautam Buddha University
Thesis Title: "An Efficient Routing Protocol for QoS Provisioning in Dynamic Multihop MANETs"
- M.Tech. in Digital Electronics (2005)
Visvesvaraya Technological University (VTU), Karnataka
- B.E. in Electronics & Communication Engineering (2003)
Visvesvaraya Technological University (VTU), Karnataka

CERTIFICATIONS & INDUSTRIAL TRAINING

- Certificate of Merit – IIT Kharagpur AI4ICPS Certificate Programme on “Hands-on Approach to AI for Real-world Applications (HAAI)” (Joint Collaboration between IIT Kharagpur and TCS iON).
- NPTEL Certification Course on “Introduction to Industry 4.0 and Industrial Internet of Things”.
- NPTEL Certification Course on “Introduction to Internet of Things”.
- Two-Week Industrial Training Programme on “Deep Learning”.
- One-Month Online Internship on “Machine Learning” organized jointly by NITK Surathkal and Pantech E-Learning.
- Participated in multiple Faculty Development Programmes (FDPs) focusing on “IoT”, “IoT with Machine Learning”, and “Recent Advancements in Machine Learning and AI”.

PATENTS PUBLISHED & GRANTED

- AI-enabled Cyber Command Console. *Application No: 475524-001 | Filing Date: 01-10-2025 | Issue Date: 07-07-2026.*
- Artificial Intelligence, Big Data, Internet of Things and Blockchain Techniques and Applications in Smart Energy Management. *Application No: 202311087301 | Filing Date: 20-12-2023 | Publication Date: 19-01-2024.*
- Internet of Things and Artificial Intelligence Based Intelligent Vehicle-Controlled Rescue System for Epilepsy Patients. *Patent Application No: 202141022696 A | Application Date: 21-05-2021 | Publication Date: 11-06-2021.*
- Method for Authenticating and Validating Quick Response Code. *Patent Application No: 202121010704 A | Application Date: 14-03-2021 | Publication Date: 09-04-2021 (Patent Office Issue No. 15/2021).*
- GPS Data Spoofing and Malfunctioning Detection System Using Classifiers. *Patent Application No: 2021100964 | Filing Date: 22-02-2021 | Status: Granted.*

BOOKS PUBLISHED

- Meena Rao, “Design, Implementation and Analysis of an Efficient Routing Protocol for QoS Provisioning in MANETs”, Advance Academic Publisher, October 2016. (ISBN-13: 978-93-84698-25-6; ISBN-10: 93-84698-25-6).
- Meena Rao, “Implementation and Analysis of File Transfer Protocol Access and Management”, Advance Academic Publisher, November 2016. (ISBN-13: 978-93-84698-26-3; ISBN-10: 93-84698-26-3).
- Meena Rao, “Development and Analysis of Automobile Recognition Device for People with Blindness”, Advance Academic Publisher, December 2016. (ISBN-13: 978-93-84698-27-0; ISBN-10: 93-84698-27-4).

BOOK CHAPTERS

- Mehra, S., Rao, M., Bansal, A. V., Rathore, N., Sidana, S., Raj, S., Sinha, A., Rao, G. M., Shamim, R., Singh, N., & Kumar, B. (2025). “Ethical Challenges and Innovations in AI-Driven Healthcare and Engineering: A Review of Blockchain, Cybersecurity, Data Privacy, and Knowledge Management”. In P. Bhattacharya, A. Hassan, H. Liu, & B. Bhushan (Eds.), *Ethical Dimensions of AI Development* (pp. 323-346). IGI Global. DOI: 10.4018/979-8-3693-4147-6.ch015.
- B. Eshwar, Harshaditya Sheoran, Shivansh Pathak, Meena Rao (2021). “Obstacle Avoidance Simulation and Real-Time Lane Detection for AI Based Self-Driving Car”, Series: *Advances in Data Engineering and Machine Learning, Advances in Data Wrangling & its Principles*, Wiley.
- Meena Rao, Richa Gupta (2021). “Impact and Suitability of Reactive Routing Protocols, Energy Efficient and AI Techniques on QoS Parameters of WANETs”, *Green Computing in Network Security: Energy Efficient Solutions for Business and Home*, CRC Press, Taylor & Francis Group.

RESEARCH PUBLICATIONS (Reputed Journals - SCI/SCIE/SCOPUS/UGC INDEXED)

- **Meena Rao**, Parul Chaudhary, Kavita Sheoran, Geetika Dhand, “A secure routing protocol using hybrid deep regression based trust evaluation and clustering for mobile adhoc network”, Peer-to-Peer Networking and Applications, Springer, Vol. 16, pp. 2794–2810, Oct 2, 2023. [SCIE, Scopus, UGC Indexed]
- Geetika Dhand, **Meena Rao**, Parul Chaudhary and Kavita Sheoran, “A Secure Routing and Malicious Node Detection in Mobile Ad Hoc Network Using Trust Value Evaluation with Improved XG Boost Mechanism”, Journal of Computer and Network Applications, Elsevier ScienceDirect, Dec 10, 2024. [SCIE, ISSN: 1084-8045]
- **Meena Rao** and Neeta Singh, “Energy Efficient QoS Aware Hierarchical KF-MAC Routing Protocol in Manet”, Wireless Personal Communications, Springer Link, Vol. 80, No. 3, pp. 1-14, Apr 12, 2018. [SCIE, Scopus, UGC-CARE Indexed, ISSN: 0929-6212]
- Richa Gupta, Deepshikha Yadav, **Meena Rao** and Ankit Gaur, “Design and Testing of SMPS in Accordance with IEC 61000-4-11 Standard with Measurement Uncertainty”, MAPAN, Springer, Vol. 37, pp. 71-80, Aug 27, 2021. [SCIE, Scopus, UGC-CARE Indexed]
- Sameer Dixit, Priyanka Pande, Shubham Sahai, **Meena Rao**, Jitendra Kumar, “Adaptive Intrusion Detection System for Wireless Sensor Networks Using Deep Learning Models”, Panamerican Mathematical Journal, Vol. 35, No. 4s, 2025. [Scopus Indexed, ISSN: 1064-9735]
- **Meena Rao** and Parul Chaudhary, “An Efficient Student Result Management System”, Journal of Technology, Vol. 12, Issue 11, pp. 690-699, Nov 2024. [UGC and Scopus Indexed, ISSN: 10123407, DOI: 18.15001/JOT.2024/V12I11.24.938]
- Parul Chaudhary and **Meena Rao**, “Blockchain, Cybersecurity and Data Privacy: A review”, Journal of Technology, Vol. 12, Issue 8, pp. 51-63, Aug 2024. [UGC and Scopus Indexed, ISSN: 10123407, DOI: 18.15001/JOT.2024/V12I8.24.589]
- **Meena Rao**, Parul Chaudhary, Amina Jafri, Mohd. Shaikhul Ashraf, “Approaches for New Product Development in Biotechnology: An Empirical Study on Problem Inventory Analysis”, African Journal of Biological Sciences, Vol. 6, Issue 6, pp. 6661-6669, July 2, 2024. [Scopus Indexed, ISSN: 2663-2187]
- Lalit Kumar Sharma, **Meena Rao**, Prabu Selvam, Anand Muley, “Role of E-CRM in effectiveness of Strategic Marketing Planning for customer Retention: An Empirical Study”, African Journal of Biological Sciences, Vol. 6, Issue 6, pp. 1606-1609, May 13, 2024. [Scopus Indexed, ISSN: 2663-2187]
- Parul Chaudhary, Rohit Sharma, **Meena Rao**, Mohit Tiwari, “Role of Machine Learning Applications in Enhancing Cyber Security Effectiveness: An Empirical Study”, Rivista Italiana di Filosofia Analitica Junior, Vol. 14, No. 1, pp. 405-411, 2023. [ESCI Indexed, ISSN: 2037-4445]
- Kavita Sheoran, Geetika Dhand, Shaily Malik, Nishtha Jatana and **Meena Rao**, "Securing the Web Applications from Cyber-Criminal Attacks", Indian Journal of Technical Education, Vol. 46, No. 1, pp. 78-81, Mar 2023. [UGC Approved, ISSN: 0971-3034]
- **Meena Rao**, Kavita Sheoran, Geetika Dhand, Shaily Malik, Nishtha Jatana, “Traversing Serverless Architecture”, Indian Journal of Technical Education, Vol. 45, No. 3, July-Sept 2022. [UGC Approved, ISSN: 0971-3034]
- Mukesh Kumar Garg, Neeta Singh and **Meena Rao**, “Design and Implementation of Trust-based Secure Routing Protocol for MANETs”, International Journal of Applied Engineering Research, Vol. 12, No. 16, pp. 5440-5446, 2017. [Scopus Indexed, ISSN: 0973-4562]
- **Meena Rao** and Neeta Singh, “Energy Efficient AODV for Backup Route Selection in MANETs”, Indian Journal of Science and Technology, Vol. 10, Issue 10, pp. 1-5, March 2017. [Web of Science, Thomson Reuters, ISSN: 0974-5645]
- **Meena Rao** and Neeta Singh, “Performance Evaluation of AODV nthBR Routing Protocol under Varying Node Density and Node Mobility for MANETs”, Indian Journal of Science and Technology, Vol. 8, Issue 17, pp. 1-9, Aug 2015. [Web of Science, Thomson Reuters, ISSN: 0974-5645]

- **Meena Rao** and Neeta Singh, “Simulation of Various QoS Parameters in a High Density MANET Set-Up using AODV nthBR Protocol for Multimedia Transmission, Data Transmission and Under Congestion Scenario”, ICTACT Journal on Communication Technology, Vol. 6, Issue 3, pp. 1155-1159, Sep 2015. [UGC Approved]

CONFERENCE PUBLICATIONS & PRESENTATIONS

- **Meena Rao**, Parul Chaudhary and Parveen Kumar, “Analyzing Airquality Index of Urban Cities and Comparing ML Models for AQI Predictions”, Presented in National Conference for Integrating Technologies, Ideas and Disciplines for Engineering Innovation (NCI-TIDE 2025), Nov 18-19, 2025.
- Parul Chaudhary, **Meena Rao** and Parveen Kumar, “Design of Digital Differentiator Using Teacher Learning Based Optimization Algorithm”, Presented in National Conference for Integrating Technologies, Ideas and Disciplines for Engineering Innovation (NCI-TIDE 2025), Nov 18-19, 2025. [Awarded Best Paper]
- Parul Chaudhary, **Meena Rao**, “Intelligent Aid: A tool for better communication and navigation for the visually challenged”, Presented in 1st IEEE International Conference on Advances in Computing, Communication and Networking (ICAC2N-2024), Dec 16-17, 2024.
- **Meena Rao**, Richa Gupta, Shivam Azad, “A study on Effective Heart Disease Prediction using various machine learning algorithms”, Presented in National Conference on Legacy of Information Communication Technology Innovations and Cybersecurity 2023, Jan 19-20, 2023, organized by USICT (Gautam Buddha University) and IETE Noida.
- Richa Gupta, Deepshikha Yadav, **Meena Rao**, Ankit Gaur, “Design & testing of SMPS in accordance with IEC-61000-4-11 standard with lower measurement uncertainty”, Presented in 7th National Conference on Advances in Metrology (AdMet 2021), Springer, MSIT, New Delhi, March 5-6, 2021.
- **Meena Rao**, “Impact of Routing Protocols and Artificial Intelligence Techniques on adhoc networks in critical scenario”, 2nd International Conference on Emerging Technologies in Data Mining and Information Security, Lecture Notes in Networks and Systems, Springer, Vol. 164, pp. 883-890, July 2-4, 2020. [Awarded Best Paper in Session Track 4: Data Science and Data Analytics | Print ISBN: 978-981-15-9773-2; Online ISBN: 978-981-15-9774-9]
- **Meena Rao** and Neeta Singh, “Performance Analysis of AODV nthBR Protocol for Multimedia Transmission under Different Traffic Conditions for Sparse and Densely Populated MANETs”, Proceedings of IEEE Sponsored International Green Computing and Internet of Things Conference (ICGCoIT), pp. 1010-1015, Oct 8-10, 2015. [IEEE ISBN: 978-1-4673-7909-0]
- **Meena Rao** and Neeta Singh, “An Improved Routing Protocol (AODV nthBR) for Efficient Routing in MANETs”, Proceedings of the 2nd International Conference on Advanced Computing Networking and Informatics (ICACNI 2014), June 24-26, 2014. Smart Innovation, Systems and Technologies, Springer, Vol. 28, pp. 215-223. [Scopus Indexed | DOI: 10.1007/978-3-319-07350-7_24 | Print ISBN: 978-3-319-07349-1; Online ISBN: 978-3-319-07350-7]
- **Meena Rao** and Neeta Singh, “Quality of Service Enhancement in MANETs with an Efficient Routing Algorithm”, Proceedings of IEEE International Advance Computing Conference (IACC 2014), pp. 381-384, Feb 21-22, 2014.

EXPERT TALKS & LECTURES DELIVERED

- Delivered an expert lecture titled “Routing Protocols for QoS Provisioning in MANETs” on June 29, 2020, as part of the Two-Week Online Expert Lecture Series on “Research Trends in ICT” (June 29 – July 9, 2020) organized by Gautam Buddha University, Greater Noida.

AWARDS, RECOGNITIONS & PROFESSIONAL SERVICE

- Best Paper Award: Awarded for the paper titled “Design of Digital Differentiator Using Teacher Learning Based Optimization Algorithm” at the National Conference for Integrating Technologies, Ideas and Disciplines for Engineering Innovation (NCI-TIDE 2025), Nov 18–19, 2025.

- Session Chair: National Conference for Integrating Technologies, Ideas and Disciplines for Engineering Innovation (NCI-TIDE 2025), Nov 18–19, 2025.
- Session Chair: 10th International Conference on Microelectronics, Computing & Communication.
- Certificate of Excellence in Research (2023): Awarded by Maharaja Surajmal Institute of Technology.
- Certificate of Excellence in Research (2022): Awarded by Maharaja Surajmal Institute of Technology.
- Letter of Appreciation (January 2024): Awarded by Maharaja Surajmal Institute of Technology.
- Letter of Appreciation (2016): Awarded by Maharaja Surajmal Institute of Technology.
- Certificate of Appreciation (Academic Year 2014–15): Awarded by Maharaja Surajmal Institute of Technology.
- Best Paper Award in Session Track 4 (Data Science and Data Analytics): Awarded at the 2nd International Conference on Emerging Technologies in Data Mining and Information Security (July 2–4, 2020).

PROFESSIONAL MEMBERSHIPS

- Life Member, Indian Society for Technical Education (ISTE)
- Distinguished Fellow Member, Indian Society for VLSI Education (ISVE)